

INTERVIEW | G. MADHAVAN NAIR

“With so many unknowns, it is like jumping into an ocean”

Minutes after the launch of Chandrayaan-1 at Sriharikota, Indian Space Research Organisation (ISRO) Chairman G. Madhavan Nair spoke to Managing Editor RAJ CHENGAPPA about its importance and India's future space missions. Excerpts:

Q. What does Chandrayaan's launch mean to India?

A. Technologically, it is a major breakthrough. This is the first time we are sending something out of the earth's gravitational field. There are so many unknowns that it is just like jumping into an ocean. We know the elementary lessons of swimming but to survive the 4,00,000-km journey from the earth to the moon would be a real challenge. It's the beginning of a new chapter.

Q. Chandrayaan is the 68th mission to the moon. Aren't we trying to reinvent the wheel?

A. The earlier missions brought back bits and pieces of information about the moon. Even NASA is sending a detailed probe next

are starting with the moon. Chandrayaan-2 has already been approved by the Government. It will have a lander and a rover that will pick up some samples and data. We are also working on a possible mission to Mars. It's the scientific temperament which we are going to kindle. Today, it's difficult to attract people towards science. We are pushing both technology and science to the frontiers. As far as space is concerned, India is a developed country. Now, we have to maintain this leadership position.

Q. What are the great benefits of doing such planetary explorations?

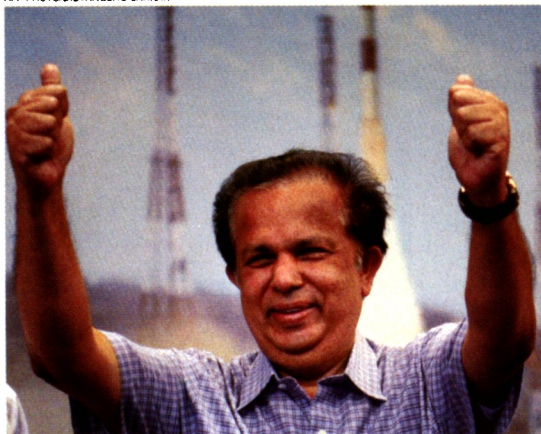
A. The agenda of space-faring nations is to explore the moon, if possible, try to exploit its resources and es-

tablish colonies—either robotic or human. And India cannot afford to lag behind in this. So we have to keep pace and as usual, we do it on a shoestring budget. The Chandrayaan mission hardly cost \$100 million which by any global standard is 20 per cent of what is usually spent for such programmes.

Q. China has just done a space walk. What are India's plans for manned space flights?

A. So far we have been concentrating on robotic explorations. But in due course, having a man behind a machine or instrument becomes important. However powerful computers you make, it cannot beat human intelligence. We have already developed a programme in which we can carry a module with two persons orbiting around the earth with our own launch vehicle so we will have a ca-

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year. This means that the knowledge of the moon is not fully available. We have geared things up in such a way that Chandrayaan shows a comprehensive map of the moon—which others have not done as yet.

Q. Isn't ISRO drifting away from its primary objective of using space for development?

A. ISRO's priority is very clear. We want to use our space-based technology and assets for enriching the quality of life for the people of our country. That will remain our objective. But a small fraction of our resources and efforts will go towards planetary explorations. We

capacity to access space. We want to achieve it before 2015. It will cost around Rs 12,000 crore and we are waiting for the Government to clear it.

Q. What is ISRO doing to improve its rocketry?

A. As far as cheap launchers are concerned, we want to maintain our leadership. Once we develop GSLV Mark 3, our next generation launcher, it will bring down the cost for a launch by 60 per cent. Even now, our launchers are highly cost-effective. So we are already competitive in the world market. But we must develop reusable launchers which will bring down the costs even further.